CNA Response to the Request for Information on Development of Artificial Intelligence (AI) Action Plan

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1. A Strategic Vision for Artificial Intelligence (AI) Leadership

The United States (U.S.) has long been a global leader in developing and perfecting advanced technologies that drive dominance in information technology, energy, manufacturing, and communications sectors. This investment has also ensured our superiority in signals intelligence (SIGINT), communications networks and technologies, and military capabilities. Today, strategic investments in AI, machine learning, and autonomous systems present opportunities to enhance all aspects of US national power—diplomatic, informational, military, and economic (DIME).

For the U.S. to maintain leadership in emerging technologies and reap economic and security benefits, a collaborative strategy involving both government and private sectors is essential. Just as government-private sector partnerships propelled advancements in the internet, global positioning systems, semiconductors, and quantum technologies, AI dominance can similarly shape the global landscape and bolster national security and U.S. economic competitiveness.

An AI Action Plan is a strategic imperative for the nation's future. It is not just a roadmap for technological advancement but a call to action for proactive leadership. The Administration must drive private sector growth, invest in research and development to bring new technologies to market, and promote public sector adoption. Achieving this requires public-private partnerships, strategic public sector investment, and AI governance standards to shape the market and ensure data security. Investing in these areas will enable the US to maintain its leadership in AI, drive economic growth, enhance national security, and promote responsible AI development.

The CNA Corporation (CNA) is eager to leverage our decades of public sector experience and capabilities to help the government achieve its goals of maintaining U.S. supremacy in AI capabilities and technology. By aiding in AI policy development, strategic planning, and implementation, CNA equips government agencies with the essential policies, procedures, and tools to seamlessly integrate AI technologies, helping enhance mission efficiency and public service effectiveness.

As a nonprofit company, CNA's ***emerging technology experts are free from the conflicts of interest inherent in companies that develop and field their own technologies***. This independence allows us to objectively assess processes, systems, and programs for our government clients and sponsors. CNA leverages data analytics and sophisticated methods to support federal, state, and local government officials as they work to protect the homeland, the American people, and industry. Our work spans the adoption of AI and emerging technologies; the resilience of U.S. supply chains; the protection of national critical infrastructure, and the increased operational effectiveness of our military in the battlespace and law enforcement responders at home.

CNA's Center for AI and Autonomy (CAIA). serves as a focal point for analytic efforts on AI, machine learning (ML), and autonomy. CAIA serves as coordination center to connect key stakeholders from the government, private sector, and academia, conducting analytics on AI adoption, use cases, prioritization, and feasibility for specific government functions. CAIA supports government decision-makers with data-driven analytics to effectively incorporate AI in support of U.S. national and economic security. Also, our CNAi2 Innovation Incubator initiative supports continuous investments in internal projects to explore new AI tools and approaches for addressing emerging challenges facing our clients. We have implemented our own internal LLM system based on governing policies compliant with current government information technology regulations and security requirements. This approach helps CNA harness the benefits of AI while managing risks and to responsibly test LLM tools and capabilities that could also prove beneficial to our clients. Our AI expertise includes:

- **Policy and Guidance Development:** CNA has collaborated with the FAA and NASA to develop classification structures and certification plans for AI technologies. We have also created maturity models to help federal agencies assess their AI capabilities and evaluated an agency's "AI readiness," particularly the availability of foundational data necessary for these systems.
- **Strategic Foresight and Planning:** CNA has facilitated strategic working groups to evaluate the risks, opportunities, and actions of emerging AI technologies. We have explored various AI applications, financial implications, and gaps in learning and governance, ensuring that both industry and the public sector are involved.
- **Technology Assessment and Implementation:** CNA has supported the development and prototyping of AI solutions for the FAA, DoD, and other entities. Our work includes establishing ML algorithm data engineering pipelines, developing AI use cases, and creating frameworks for AI prioritization and certification.

CNA stands ready to support the Government's efforts related to AI and offers the following ideas and suggestions for consideration.

2. AI Focused Partnerships with the Private Sector

CNA suggests that new models for supporting and benefiting from the private sector is essential for maintaining U.S. supremacy in AI. Traditional public-private partnerships (PPPs), as well as alternative models such as joint ventures, alternative financing structures, and the development of innovation funds or financing are essential for maintaining U.S. supremacy in AI. Through these collaborations the strengths of both sectors are leveraged to accelerate innovation, share resources, and address complex challenges. The private sector brings agility, cutting-edge technology, and significant investment in AI research and development, while the public sector provides strategic direction, regulatory oversight, and funding for foundational research. By working together, these partnerships can drive advancements in AI that are critical for national and homeland security.

To foster effective public-private partnerships, CNA recommends that AI action plan specifically consider how to: 1) invest and/or motivate investment in U.S.-based AI technology development; 2) drive investment in U.S. firms from our strategic allies; and 3) put in place policies that drive U.S. growth and AI dominance in the global AI market. The first step, enhancing federal funding, is crucial. Support for many types of collaborations (as outlined below) can help forge trusted partnerships to address challenging issues in support of the public interest. Continued support for establishing regulatory sandboxes is also important. ***This allows AI companies to test new technologies in a controlled environment without the full weight of regulatory constraints***, helping to identify potential risks and benefits early in the development process. Additionally, developing secure data-sharing frameworks can facilitate access to high-quality datasets for AI research while protecting privacy and sensitive information, encouraging collaboration between public and private entities.

Talent development programs are also essential. Implementing programs that support STEM education for students pursuing AI-related fields, as well as creating pathways for AI experts to move between the public and private sectors, can help share knowledge and expertise. Furthermore, encouraging the formation of public-private research consortia can bring together government agencies, academic institutions, and private companies to work on AI projects of mutual interest, pooling resources and expertise to tackle complex challenges.

CNA's history of testing, evaluating, and identifying new technology for the Department of Defense, has always included strong connections across government, industry, and academia. CNA's technologists are interdisciplinary, bringing expertise in computer science, operations research, engineering, and policy analysis. Examples of CNA-driven collaboration methods conducted for our federal clients is summarized Table 1 below. This presents only a sample of options available to the government in this area.

Table 1. Collaboration Methods to Engage Diverse Stakeholders

Collaboration Method	Description	Benefits
Strategic Foresight and Horizon Scanning	**Convening Experts:** Bringing together experts from industry, academia, and govt. to discuss and analyze future trends in AI. **Best Practices:** Creating processes for identifying and analyzing the impact of emerging AI technologies on various sectors.	**Future Trends Identification:** Helps stakeholders anticipate and prepare for future AI trends. **Informed Recommendations:** Provides data-driven recommendations for strategic planning. **Enhanced Preparedness:** Ensures stakeholders are ready for emerging AI impacts.

Collaboration Method	Description	Benefits
Facilitation and Decision Support	**Decision-Making Support:** Facilitating decision-making for government agencies with comprehensive analysis and recommendations. **Technology Assessment:** Evaluating the impacts and benefits of AI technologies. **Stakeholder Engagement:** Leveraging connections with non-profits, academia, and industry to bring diverse perspectives into the decision-making process. **AI Policy and Guidance Development:** Assisting in the development of AI policies and guidelines.	**Enhanced Decision-Making:** Provides agencies with the necessary information and analysis to make informed decisions. **Effective AI Implementation:** Ensures AI technologies are implemented responsibly and effectively. **Bridging Gaps:** Facilitates collaboration between public and private stakeholders to achieve common goals.
Advisory Boards and Stakeholder Engagement	**Advisory Boards:** They include representatives from industry, government, and civil society to foster collaboration and information sharing. **Open Communication Channels:** Creating open channels of communication between different stakeholders to ensure transparency and mutual understanding. **Domain-Specific Research:** Engaging in research organizations for joint R&D and best practices development for AI innovation.	**Promoted Collaboration:** Forms trusted relationships among diverse stakeholders. **Best Practices Development:** Helps develop and disseminate best practices for AI innovation. **Diverse Perspectives:** Ensures that AI initiatives consider and incorporate diverse perspectives and expertise.
Gaming and Exercises	**Exploring AI Implementation:** Using gaming and simulation exercises to explore the practical implementation of AI technologies in various scenarios. **Decision-Maker Engagement:** Bringing together diverse sets of decision-makers to discuss and evaluate the operational, logistical, and strategic implications of AI integration. **Sector-Specific Evaluation:** Assessing the potential impact and integration of AI in sectors such as public safety and national security.	**Practical Insights:** Provides stakeholders with practical insights into the challenges and opportunities of AI implementation. **Strategic Planning:** Enhances strategic planning and decision-making through simulated scenarios. **Challenge Identification:** Identifies potential challenges and solutions for AI integration in specific sectors.

Following, please find specific examples of CNA's AI collaborations with public sector entities and the associated outcomes and benefits.

Example 1: Horizon Scanning. CNA supports strategic foresight efforts at the Department of Homeland Security and the National Academy of Sciences by convening expert stakeholders from industry, academia, and government to identify processes and best practices around the emerging trends in AI that impact various infrastructure sectors. This approach provides our clients with informed recommendations for strategic planning and enhances preparedness for emerging AI impacts. Further, CNA has also engaged law enforcement practitioners and researchers to evaluating the opportunities, actions, and risks of emerging technology/AI in corrections and policing. Using a futures analysis framework, we have facilitated strategic working groups focused on examining the world in 2040. Our experts have explored issues such as the various applications of AI technologies, financial implications for already strained government agencies, opportunities for improving the health and safety of incarcerated individuals and staff, and gaps in learning and governance related to AI.

Example 2: Gaming and Exercises. CNA has explored AI implementation and futures in a variety of games and exercises. For example, our gamers have brought decision- makers together to explore the operational, logistical, and strategic implications of AI for "intelligentized warfare" and for current and future unmanned systems. Our gaming approach for evaluating AI implementation is adaptable and can be used to examine a range of topics from AI integration in public safety, to AI integration with other systems, to adversary adoption of AI, and the use of AI in areas of instability or conflict. Additional information on our gaming capabilities can be found on the [CNA Website](#).

Example 3: Advisory Boards: In support of AI Action Plan implementation, the government may consider leveraging and maintaining strategic partnerships through the establishment of an advisory board that guides initiatives and policy development. Effective advisory boards consist of key spokespersons representing the initiatives partner organizations, public, and private sectors, and the user community. Specifically, advisory boards should also include those who have expertise in the topic at hand (i.e. AI hardware, software and implementation and its optimization). A Federal AI Advisory Council, or similar, could include members such as:

- Industry and Technology Leaders (Google DeepMind, NVIDIA, Shell AI, GridX),
- Department AI Officers, Commerce, State, and other federal agencies (ARPA-E, DoD),
- University AI and Energy Research Centers (MIT energy Initiative, Carnegie Mellon AI Lab),
- Civil Society and Policy Experts

An advisory board would help drive cohesion in the development of the initiative while fostering collaboration among the industry and civil society groups in which they are embedded. In the past, CNA has brought these groups together to support the government with decisions ranging from the organizational design of US Space Force, the evaluation of the US information operations posture, and to identify approaches for stronger community preparedness and emergency response. All these advisory boards open channels of communication between industry, government, and society that create connections between initiative priorities and additional engagement opportunities.

3. AI and National Security

Maintaining supremacy in AI is critically important for the United States to ensure national security and global leadership. AI technologies provide a strategic advantage in military operations, enhance defense capabilities, and act as a force multiplier by increasing the efficiency and effectiveness of military personnel and equipment. By staying at the forefront of AI innovation, the U.S. can deter adversaries, protect its critical infrastructure, and respond swiftly to emerging threats. ***Furthermore, AI leadership enables the US to set global standards and norms promoting U.S. leadership worldwide.***

From a national security perspective, the ability to leverage AI for intelligence, surveillance, reconnaissance, cybersecurity, and autonomous systems is essential for maintaining a competitive edge over potential adversaries. Investment in AI research and development, talent cultivation, and robust cybersecurity measures are key to sustaining this dominance. By implementing policies that support these areas, the U.S. can ensure that it remains at the cutting edge of AI technology, thereby safeguarding national security and reinforcing its position as a global leader in innovation and defense.

Application Areas. There are myriad application areas that are primed for AI breakthroughs across the military, science and applied energy that could have national security implications. From an implementation perspective, the AI Action Plan should support government and industry to effectively realize the breakthroughs. Following are some specific areas of potential breakthrough where CNA is already embedded with leadership and industry.

> **AI Prioritization Use Case.** CNA's scientists are supporting the US Space Force (USSF) with an evaluation of how they could develop and apply AI applications across the service, given the current technical and structural limitations of these technologies. We created a "little-picture" framework for prioritization at the mission level and a "big-picture" framework for prioritization at the service level. We also identified how USSF might approach a service-level AI strategy and how the components of the AI prioritization frameworks would inform that strategy.

CNA has supported development of AI algorithms for real-time analysis of video feeds from drones and satellites that can help create models to enhance object detection and tracking capabilities, improving situational awareness for military operations. CNA has also utilized Natural Language Processing (NLP) techniques to analyze and interpret large volumes of intelligence data. Projects have focused on improving the extraction of relevant information from diverse sources, enabling more efficient and accurate threat assessments. CNA's supply chain applications in the AI arena include a deeper understanding of freight movement optimization and risk assessment to characterize patterns in vehicle flows and unlock strategies to alleviate bottlenecks while mitigating risks to communities or critical infrastructure. The ability to better characterize trucking and shipping involved in freight movement and cargo operations would help the US optimize global trade flows seasonally, reduce accidents and congestion, and reroute traffic in case of operational disruptions ranging from accidents to piracy and acts of war. The magnitude of data and model training required to facilitate work in this area would require the resources that the Action Plan can detail and support.

In the applied energy sector, aspects of fuel demand estimation and supply availability could benefit from AI to help monitor daily fuel usage at multiple spatial and temporal scales. Relatedly, fuel rack congestion monitoring through AI technologies can identify when truck racks at fuel terminals are experiencing long lines, while new facility capacity planning applications could identify where additional capacity for renewable diesel generation would be most feasible. Our development and implementation of the [Supply Chain Analysis Network](#) (SCAN) has increased our visibility and work on where AI could be integrated into the Energy supply in particular, and we would be happy to provide more information in this domain.

Science applications primed for breakthroughs include the characterization of cropland productivity and price forecasting where AI would help evaluate crop production and resulting market prices for key commodity crops to inform agricultural policy, international trade, water management, and biofuels/energy policy. Water stress and competition models are also primed for breakthroughs. They can be used to understand the nexus of federal and state water rights and management controls that are complicated and often prone to significant legal battles during times of water shortages. To that extent, AI applications could aid in understanding pricing or management strategies' likely effects on water prices and resulting conservation behaviors.

4. Governance of AI Responsibilities

To maintain the U.S.'s global leadership in AI, AI governance must be both structured and adaptable. This approach will support ongoing AI innovations while being responsive to new challenges. One consideration to ensure consistent and coordinated government oversight of AI activities is the organizational structure for this function. Questions like whether a single Department, or a disaggregated structure would more successfully lead to streamlined policy development and implementation? And, which organizational structures would ensure a cohesive and strategic approach to maintaining AI dominance?

Current Structure. As noted in **Table 2**, current AI responsibilities are shared across multiple government organizations. The National AI Initiative Office (NAIIO), established by President Trump, coordinates federal efforts in AI research and policy. The NAIIO plays a pivotal role in aligning various federal agencies' AI activities, promoting research and development, and ensuring the US maintains its leadership in AI technologies.

At one end of the organizational spectrum, the NAIIO could be elevated to a formal Bureau or Department, its roles and responsibilities significantly strengthened, potentially bringing greater efficiency and authority to AI governance. As a Bureau or Department, it could consolidate AI oversight, lead collaborations, eliminate regulatory fragmentation across multiple agencies, and ensure a cohesive national AI strategy. With added authority, it could set enforceable AI standards, conduct risk assessments, and certify AI systems for safety, ethics, and transparency.

A Bureau or Department structure could also streamline compliance for businesses by offering a single point of contact for AI regulations while fostering industry-driven innovation through public-private collaboration. Additionally, it could play a central role in securing AI supply chains, mitigating AI risks, and aligning US AI leadership with global competitors. By bringing AI governance under one roof, the US could respond more quickly to emerging AI challenges, balance regulation with innovation, and maintain its competitive edge in AI technologies.

At the other end of the spectrum is a disaggregated organizational design, which allows for autonomy in decision-making on AI, and potentially more clarity on the feasibility of specific AI use cases. This disaggregated model can allow for systems to be developed that meet the direct and specific needs of a sub-organization. It can also distribute risk through a more diversified approach. This can be effective if the disaggregated organizational design also has a strong governance across agencies, bureaus, and offices to ensure consistency and standards.

Table 2. Government-wide AI Responsibilities

Agency/Organization	Primary AI Responsibilities
National AI Initiative Office (NAIIO)	Coordinates federal AI efforts and policy across agencies.
National Institute of Standards and Technology (NIST)	Develops AI standards, risk frameworks, and safety guidelines.
White House Office of Science and Technology Policy (OSTP)	Shapes national AI strategy, ethics, and R&D priorities.
Department of Commerce (DOC)	Oversees AI's economic impact, trade policies, and industry collaboration.
Federal Trade Commission (FTC)	Regulates AI in consumer protection, data privacy, and fair competition.
Department of Defense (DoD) / Joint AI Center (JAIC)	Develops and integrates AI in national security and military applications.
Department of Energy (DOE)	Advances AI for energy research, grid management, and climate solutions.
National Security Commission on AI (NSCAI)	Advises on AI for defense, cybersecurity, and national security.
Food and Drug Administration (FDA)	Regulates AI in healthcare, medical devices, and diagnostics.
Securities and Exchange Commission (SEC)	Monitors AI use in financial markets, fraud detection, and trading algorithms.
U.S. Patent and Trademark Office (USPTO)	Handles AI-related patents and intellectual property rights.
Department of Homeland Security (DHS)	Uses AI for cybersecurity, border security, and threat detection.

Need for a Unified Strategy. Regardless of where on the organizational spectrum AI decision-making occurs, it is critical that a unified national strategy for AI, spearheaded by the AI Action Plan is in place. With an AI Action Plan in place, the goals and priorities of various governmental departments around AI will be aligned. The AI Action plan should drive the organizational design of the function, the harmonizing of regulations across different sectors, and the avoidance of overlaps and contradictory rules. It would ensure that clear guidelines and frameworks for AI development and deployment are in place, and all agencies follow the same standards. Importantly, the AI Action Plan will provide a roadmap to resolve any conflicts that arise between different regulatory bodies regarding AI policies.

Because AI is evolving rapidly, a unified national strategy for AI could spearhead and promote policies and regulations to manage responsible AI diffusion through effective export control policies, as currently being defined in the DOC's Framework for AI Diffusion. It could provide national security oversight, including threat assessments to monitor global AI advancements and identify potential threats to national security, particularly from strategic competitors like China. It can facilitate collaboration across intelligence agencies to analyze international AI developments and their implications for U.S. security and support defense integration to ensure the U.S. defense apparatus incorporates cutting-edge AI technologies to maintain a strategic advantage.

A successful recent example of a new organizational structure put in place to drive innovation was the establishment of U.S. Space Force. With a focus on innovation in space and new risks emerging in the space domain, U.S. Space Force represented the consolidation of defense space related missions. CNA played a pivotal role in the establishment of the foundation for the U.S. Space Force by developing an organizational design for creating the separate military department. CNA's approach included an Operational Environment Analysis, a Department Design for the new department and its subordinate divisions, and Risk Analysis and Legislative Modifications detailing the impacts on existing departments and the resources needed for its functions.

Drawing parallels from the successful consolidation of space-related missions for U.S. Space Force, a similar approach could be used to structure AI-related responsibilities. The goal would be to enhance the government's ability to achieve its goals of maintaining US supremacy in AI. The kinds of questions that would drive the approach would include (among others):

1. **Centralized or Disaggregated Coordination**: Which organizational structure promotes efficient AI initiatives across various government sectors, ensuring a unified strategy and efficient resource allocation.
2. **Stakeholder Engagement**: How to maximize engagement with industry, academic, and military stakeholders to gather insights and foster collaboration, like the approach taken for the U.S Space Force?
3. **Operational Environment and Gap Analysis**: Conducting a thorough analysis of the current AI landscape, identifying gaps, and proposing solutions to address these gaps.
4. **Legislative and Policy Framework**: Developing a robust legislative and policy framework to support AI initiatives, ensuring compliance with ethical standards and international regulations.
5. **Risk Management**: Implementing a comprehensive risk management strategy to address potential challenges and mitigate risks associated with AI development and deployment.

Of course, creating a new entity that either aggregates or disaggregated responsibilities would require strong leadership to navigate and address possible bureaucratic complexities to include Congressional approvals, buy in from industry, and agency turf wars.

CNA's organizational design and AI technologists stand ready to assist the government in building an effective AI oversight capability, driving innovation and maintaining a competitive edge in the global AI landscape.

5. AI Research and Development (R&D)

R&D is crucial for the continuous innovation needed to stay ahead in the rapidly evolving field of AI. It helps develop new algorithms, models, and applications that push the boundaries of what AI can achieve. Investing in R&D ensures that the US remains at the forefront of technological advancements, enabling the development of cutting-edge AI technologies that can be leveraged across various sectors. Most critically, federal R&D funds can provide the upfront investment necessary for emergence of new U.S. based technologies to enter and grow in the market.

Domestic benefits of R&D activities is the creation of high-skilled jobs and a U.S. competitive workforce. This contributes to economic growth and positions the US as a leader in the global AI market. Continuous R&D efforts help maintain the competitiveness of US industries by integrating AI into products and services, enhancing their value and global market appeal. Advanced AI technologies developed through R&D can enhance national security by improving defense systems, intelligence analysis, and operational efficiencies.

To truly support a competitive hardware ecosystem, the Action Plan could include additional investments to track, analyze, and explore broader trends, barriers, and opportunities influencing the trajectory of AI compute, and policy actions necessary to minimize future risks to American leadership. That may take the shape of partnership with organizations like CNA that have a history of working in the public-private partnership space with expertise in identifying and analyzing emerging AI trends and technologies that could have major implications for AI competitiveness and readiness. This would include the tracking of domestic manufacturing capacity of advanced chips, securing underlying supply chains, understanding the potential ramifications of increasing adoption of open-source instruction sets (e.g., RISC-V), investing in and tracking progress in non-silicon-based chips and hybrid systems, supporting efforts to address power availability challenges, and advancing packaging research and production.

Leveraging government grants and contracts. The Action Plan should emphasize new and expanded funding mechanisms including additional grants, contracts and new work under existing contracts targeting emerging AI technologies and to promote ongoing innovation in the field. In this context, leveraging the expertise of nonprofits can significantly enhance the effectiveness of AI initiatives.

AI Technology Acquisitions. CNA's scientists and analysts have, for decades, provided technical assessment and support to our clients around small- and large-scale technology buying/acquisitions. At the early stages, we conduct Analyses of Alternatives to provide decision- makers with a data-driven understanding of capability needs. Then, throughout the acquisition lifecycle, CNA provides analyses of return on investment; impact on manning, training, and education as new technologies are brought online; and continuous test and evaluation to ensure that new technologies are best meeting the needs of the government.

Nonprofit organizations like CNA are designed to support long-term analytic and development needs and offer several unique advantages to the government, particularly when addressing complex challenges and proposing effective solutions for AI integration into agency mission-critical operations. Nonprofit organizations operate with a high degree of independence and objectivity, ensuring that their recommendations are unbiased. Unlike traditional contractors, nonprofits are not profit-driven and focus solely on the public interest and the government's priority missions and goals. As a nonprofit, CNA does not have financial stakes in AI technologies. This independence ensures that our analysis and recommendations are free from commercial bias, providing government clients with impartial advice. Our evaluations of AI processes, systems, and programs are based solely on their merits and alignment with government objectives, leading to more accurate and trustworthy insights. This enables us to develop trusted, long-term relationships with government organizations, fostering deep institutional knowledge and continuity. This is particularly beneficial for complex and evolving fields like AI, where ongoing collaboration and understanding of agency-specific needs are crucial.

For example, drawing on our extensive 80-year history of supporting the DoD, CNA has developed a unique capability to provide in-depth and reliable support for Navy AI efforts. This includes exploring new AI technologies, algorithms, and applications that can be integrated into Navy operations. Our research helps identify potential AI solutions for various challenges, from logistics and maintenance to combat systems and intelligence analysis. This includes testing AI systems in real-world scenarios to validate their performance and identify potential improvements.

CNA also fosters interdisciplinary collaboration by bringing together experts from various fields, including computer science, engineering, operations research, and social sciences. This collaborative approach ensures that AI solutions are comprehensive and consider multiple perspectives, enhancing their effectiveness and applicability.

Prize Based Challenges. Prize-based challenges or competitions are another mechanism used to drive innovation in AI technologies and encourage industry to develop AI- based technologies and solutions to support mission critical services that serve the public interest. The Action Plan should support innovation through encouraging and supporting competitions managed by specific government organizations. For example, the recent NIST Artificial Intelligence for Industrial Internet (AI3) Challenge sought AI solutions to support First Responders exchanging data and providing situational awareness in emergency situations. CNA developed a system that uses AI to ingest, fuse, classify, and analyze known and unknown IoT sensor data into an intuitive data visualization interface for first responders and unit commanders.

We suggest structuring the challenges in stages that move through concept, prototype, and live demo. Important to keep in mind when developing a prize-based competition is how the goal of the competition is formulated and published, the use cases provided to competitors, the system requirements, and overall scoring criteria. These bounding criteria should be developed to specifically support strategic priorities and the goals to drive innovation.

6. Summary

CNA is honored to contribute recommendations for an AI Action Plan that will secure the United States' continued dominance in artificial intelligence. AI is not just a technological advancement; it is a critical driver of economic growth and national security, with limitless potential to transform various sectors and maintain American exceptionalism. A robust federal AI governance structure is essential for coordinating efforts and fostering groundbreaking AI innovations and applications.

Supporting R&D activities in AI is vital for driving innovation and keeping the U.S. at the forefront of technological advancements. This investment ensures that America remains the global leader in AI, capable of setting standards and influencing the international landscape. Furthermore, fostering public-private partnerships is crucial to ensuring a diverse range of stakeholders contribute to and benefit from AI innovation, creating a collaborative ecosystem that accelerates progress and addresses complex challenges.

CNA stands ready to support the government's AI plans and strategies, providing comprehensive analysis and solutions to ensure the United States remains at the pinnacle of AI technology and its applications. We look forward to continuing our support for the government's efforts to harness the transformative power of AI to drive American exceptionalism and global leadership.